

Obhoy: A Multi-Line Blockchain Claims-Integrity Protocol for Low-Trust Insurance Markets

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Abstract—Insurance fails in low-trust markets in large part because of how the record is kept, not only because of how the risk is priced. The failure is not confined to one line of business. In Bangladesh, general insurers settled 9.37% of filed claims in the final quarter of 2025 [1]. Crop-insurance research records farmers calling insurance a “Ponzi scheme” [2]. In 2018 the state made third-party motor cover *optional*, dropping a requirement it could not enforce [3]. Rich markets price the same problem rather than solve it, at an estimated \$308.6 billion a year in the United States [4]. This paper argues these are one problem: nobody can independently check that the insured event happened, because the only record comes from a party with a stake in it. One piece of infrastructure answers all of them. Obhoy is a permissioned claims-integrity protocol on which no insurer, provider or intermediary holds the complete record of a claim. It defines the insurable event as a *single-use, multi-party-attested on-chain asset*, and closes four separate verification failures. An attestation quorum spans classes with opposed interests. An event-uniqueness invariant, held apart from entitlement uniqueness, lets genuine dual cover pay while stopping the same loss being recovered twice across insurers who cannot see each other. Parametric or defined-benefit settlement breaks the link between payout and a claimant-supplied invoice. Public anchoring lets a buyer check an insurer’s settlement record before buying. One design serves every line — a hospital stay, a failed harvest, a collision and a fire differ in three parameters — though not with equal force. Healthcare is the first implementation priority: the context chosen to validate the protocol, not the scope of it. The rollout follows two precedents: B3i, a twenty-carrier consortium that raised \$23m and closed in 2022 for want of volume [5], and Insurwave, which reached 1,000+ vessels by anchoring on one committed client [6].

Index Terms—claims integrity, multi-line insurance, low-trust markets, Hyperledger Fabric, parametric insurance, asset tokenisation, consortium governance, SDG 1, SDG 3.8

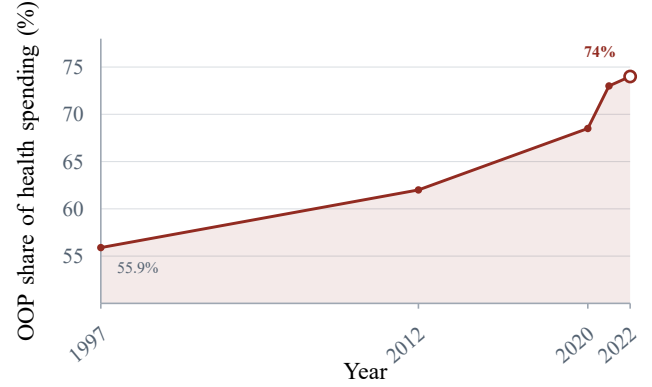


Fig. 1. Out-of-pocket share of total health expenditure, Bangladesh, 1997–2022. The trajectory is not stable; it is worsening.

Source: 6th Bangladesh National Health Accounts, Health Economics Unit, MoHFW [7]; BIDS analysis of HIES 2022 [8], [11].

I. INTRODUCTION

Three facts about insurance in Bangladesh are usually told as three separate stories.

The first is health. Households pay for illness in cash. Out-of-pocket spending rose from 55.9% of total health spending in 1997 to about 74% by 2022 (Fig. 1). Only Afghanistan is higher in the region [7], [8], [9]. A single hospital stay is among the country’s largest causes of poverty.

The second is agriculture. Sadharan Bima Corporation and the Bangladesh Meteorological Department ran a weather-index crop insurance pilot. It enrolled 6,772 marginal farmers across three districts and collected premiums through bKash [10]. It did not scale. The research records why: farmers called insurance a “Ponzi scheme”, and reported that financial firms had defrauded them before [2].

The third is motor. In 2018 the Road Transport Act made third-party cover optional [3]. The state dropped a requirement it could not enforce, and set no penalty for

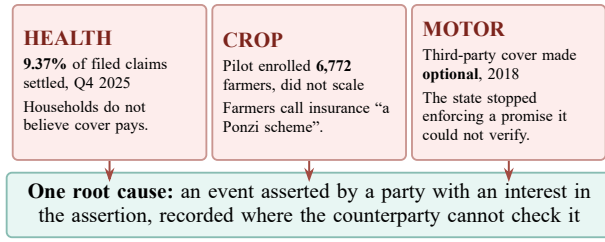


Fig. 2. Three lines, three symptoms, one failure. The trust deficit in Bangladeshi insurance is not a property of health cover; it recurs wherever the insured event is asserted rather than independently verified.
Sources: settlement ratio [1]; crop pilot enrolment [10] and farmer attitudes [2]; motor liability [3].

ignoring it.

These are one story (Fig. 2). In each line the product is a promise about a future event. In each line the party who says the event happened is the party paid for saying so. In each line the record sits with someone the other side has reason to distrust. A promise nobody can check does not get bought. When it is bought and then broken, nothing records the breach where the next buyer could see it.

This is not only a developing-market problem. The Coalition Against Insurance Fraud puts US insurance fraud at \$308.6 billion a year, across life (\$74.7bn), Medicare and Medicaid (\$68.7bn) and property and casualty (\$45bn) [4]. Rich markets do not lack verification for want of technology. They price the loss into the premium instead. Bangladesh cannot: the loading costs more than the customer can pay, so the market never forms.

A. The argument of this paper

We make three claims.

- 1) The failure splits into **four separate verification failures** (Section III), and the same four recur in every line of business.
- 2) All four have one answer: a shared claims record that no participant controls, on which the insured event becomes a single-use asset attested by parties with opposed interests (Section IV).
- 3) The design does not depend on the line. A hospital stay, a failed harvest, a collision and a fire differ in detail but share one shape. One protocol can serve several markets while being deployed in one at a time.

B. Contributions

The main contribution is a **multi-line claims-integrity protocol for low-trust insurance markets**. Within it, the paper contributes:

- 1) The *insurable event* as a single-use, multi-party-attested on-chain asset, with worked settings for health, crop, motor and property (Section IV-B);
- 2) A *multi-class attestation quorum* written as a Fabric endorsement policy, so it sits at protocol level and application code cannot bypass it. It makes payment conditional on independent attestation; it does not establish that the event physically occurred (bounds Failure 1);
- 3) An *event-uniqueness invariant* that refuses a duplicate at commit rather than catching it after payment. We hold it apart from *entitlement uniqueness*, so genuine dual cover is coordinated instead of blocked, and we state the identity assumption it rests on (closes Failure 2, Sections IV-C and V-C);
- 4) A *settlement basis independent of claimant-supplied evidence*: parametric where an index exists, defined-benefit otherwise. Inflating a document then earns nothing (closes Failure 3); and
- 5) A *publicly anchored transparency ledger*, so a buyer can check an insurer's settlement record before buying, with decentralisation measured rather than asserted (closes Failure 4, Section VI).

The deployment plan is drawn from two precedents rather than from assertion. B3i had twenty carriers behind it and wound up for want of volume. Insurwave reached scale by anchoring on one committed client (Section XII-B).

II. OBJECTIVE

Obhoy pursues three goals that lock together.

1. Make cover affordable by taking fraud out of the premium. In a market where claims cannot be checked, premiums must be loaded to absorb fraud, duplication and inflation [4] — a loading Bangladesh cannot pay. The objective is to drive the cost of verifying a claim toward zero, so the premium can fall toward the fair price of the risk itself.

2. Make the insurer's promise checkable. Obhoy publishes, per pool and per period, how many claims were received, settled and denied, and why. The record is append-only and the insurer cannot revise it. A buyer can check an insurer's behaviour before buying, and a regulator can check it continuously instead of once a year.

3. Build shared verification infrastructure that no participant owns. Obhoy is deliberately not an insurance company. It is the layer on which licensed insurers, community mutuals, takaful operators and government

schemes all settle claims against one common record, while staying commercial competitors.

III. PROBLEM

A. Four verification failures

Insurance in a low-trust market fails because four verification failures feed each other (Fig. 4). Each one is solvable alone. Together, without shared infrastructure, they are not. None of them is specific to a line of business.

Failure 1 — the insurer cannot verify that the event occurred. The claim arrives with evidence written by the party being paid: an invoice and discharge summary from the hospital, a damage assessment from the garage, a loss declaration from the farmer. Nothing independent confirms it.

Failure 2 — no insurer can see another insurer's claims. The same loss can go to two carriers and neither will know. This cuts both ways. Duplicate recovery cannot be caught, and two valid policies cannot be coordinated either. Claims data is a competitive asset. No insurer will hand it to a rival, and none will accept a rival as its keeper.

Failure 3 — claimant-supplied evidence is the only measure of loss. When payout tracks a document the claimant or provider writes, every incentive points at inflating it. Policing that needs assessment capacity nobody has at scale.

Failure 4 — the policyholder cannot verify the insurer. This is not hypothetical. In 2024 Bangladeshi insurers received Tk 16,484 crore in claims and settled Tk 9,476 crore, a 57% ratio [12]. By the final quarter of 2025 they settled Tk 372 crore of Tk 3,971 crore filed — 9.37% — leaving Tk 6,934 crore unsettled across the industry [1]. This is not one bad quarter. Fig. 3 plots the sector's own reported ratio over six periods. The belief that insurers do not pay is closer to a documented fact than to a perception problem. **A household cannot check it before buying**, because no verifiable insurer-level record exists in a form a family can read.

The average hides the worst of it. Table I names four insurers whose conduct is already public record [14].

B. The same four failures in every line

The four failures above never mention health, because they do not depend on it. The lines differ in their economics. They break at the same joint: the moment somebody must establish that an event happened before money moves. In health the hospital being paid says the admission happened. In crop the farmer claiming says

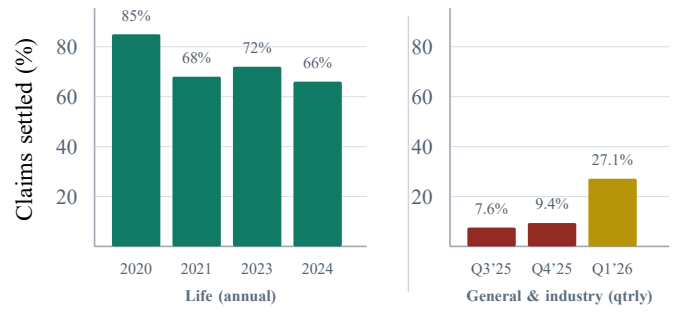


Fig. 3. Claims settlement ratio, Bangladesh insurance sector. **Left:** life segment, annual, 2020–2024. **Right:** general insurance and industry total, quarterly, Q3 2025–Q1 2026. Two panels, not one timeline: the segments differ in reporting cadence and are not comparable period-for-period.

Source: IDRA data as reported by *The Daily Star* and *The Business Standard* [13], [12], [1].

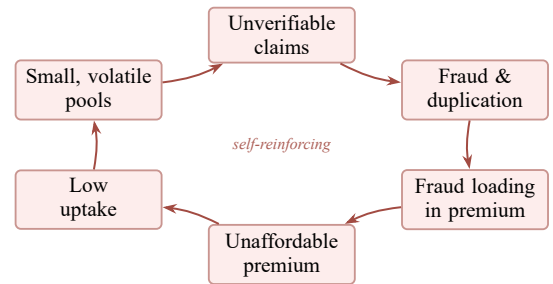


Fig. 4. The reinforcing failure loop. Each stage causes the next, and the sixth returns to the first — which is why no single intervention breaks it.

TABLE I
NAMED INSURERS AND DOCUMENTED CLAIMS-SETTLEMENT FAILURES

Insurer	Settled	What happened
Sadharan Bima Corporation (state-owned)	3.41%	Tk 2,187 cr left unresolved; mandatory insurer for public-sector assets
Green Delta Insurance	3.67%	Declared a 27% cash dividend to shareholders in the same period
Fareast Islami Life Insurance	6.59%	Directors embezzled an estimated Tk 2,100–2,367 cr (2010–2020), per a joint IDRA/BSEC audit
Padma Islami Life Insurance	4.49%	Show-cause notice over Tk 211 cr unsettled; now selling fixed assets to cover it [15], [16]

the yield failed. In motor the garage quoting the repair says how bad the damage is. In property the policyholder and their own assessor say what burned. Every time, the claim starts with the party whose income depends on it,

	P1	P2	P3
Per-insurer databases (status quo)	○	○	○
Regulator registry + audit logging	◐	○	○
Witnessed verifiable log (CT-style)	●	○	○
Permissioned DLT (Obhoy)	●	●	●

● holds ◐ partial ○ does not hold

Fig. 5. Four architectures for a shared claims record. **P1**: a recorded fact cannot be silently changed later. **P2**: an invalid write is refused *at commit* by parties with opposed interests. **P3**: the operator cannot silently leave a record out. A witnessed transparency log [17] already wins P1, so the case for a ledger rests entirely on the two columns to its right.

and no counterparty can check it.

US fraud spreads across lines the same way (Section I) [4]: fraud is not concentrated in one line, because the verification gap is not either.

C. Why a ledger — argued against the strongest alternative

A ledger is only justified where a simpler design fails. The fair comparison is not one insurer’s database, which fails at once. It is the best design available *without* a ledger: a regulator-run registry published as an append-only **verifiable log**, in the manner of Certificate Transparency. Independent witnesses gossip signed tree heads, so no operator can show one history to one party and a different history to another [17]. That design is strong. It is cheaper to run than consensus and familiar to regulators.

Fig. 5 scores four architectures on the three properties this problem needs. Each property can be falsified on its own.

The log gives P1, and P1 is most of what “blockchain” is usually asked to supply. Suppose Obhoy needed only that a published settlement ratio cannot be rewritten later (Failure 4). Then a witnessed Merkle log would be the right answer, and much cheaper than Fabric. We do not claim otherwise.

What it cannot give decides the question. A transparency log records the writes accepted, not which writes *should* have been. The uniqueness invariant of Section IV-B must run at commit time, against state held across insurers, because a duplicate caught after settlement is money already gone — and on a single-operator log, whoever runs that check is exactly the party with an interest in the answer. A Fabric endorsement policy makes the check multi-party instead: settlement

is invalid unless independent peers, one of them unpaid, ran the same invariant and agreed (P2). A log also proves only what it contains, never what it omits; an operator can decline to record an awkward claim and no audit shows the gap (P3). The write rule itself needs governing too: one operator can amend a log’s rule alone, where a chaincode change needs the multi-class approval of Section VI.

The concession. If IDRA ran a witnessed verifiable log and required insurers to use it, Failure 4 would be largely solved and part of Failure 2 with it — a good outcome, not a threat. Obhoy’s data model ports to such a log unchanged, and the case for a ledger narrows to P2 and P3. What we reject is the weaker claim that a replicated database with audit logging is enough: there, the operator who can be pressured is also the operator who writes the audit log.

D. What is out of scope

Obhoy does not judge whether treatment or repair was clinically or technically appropriate. It does not replace records systems and it does not underwrite risk. It establishes **that an event happened, that it happened once, that it falls in a covered category, and whether payment followed.** Everything past that belongs to licensed insurers, domain professionals, and the appeals process of Section VI.

IV. SOLUTION

A. The claim, stated once

Obhoy is a shared claims-integrity layer on which no single insurer, provider or intermediary holds the complete record of a claim. That is the proposal. Everything below is an instance of it. The cycle it secures is domain-independent: event, assertion, independent corroboration, refusal of conflicts at commit, eligibility, rule-bound settlement, auditable history. Domains differ in evidence and parameters, not in that sequence.

The four failures of Section III are separate, and each gets one mechanism (Fig. 6). Read the project that way. Event uniqueness is the newest of the mechanisms, but it is a mechanism, not the thesis. A system that only stopped duplicate claiming would leave Failures 1, 3 and 4 open, and those are the ones that stop buyers believing the promise at all.

B. The primitive: an insurable event as a single-use asset

The protocol defines one object. An **insurable event** is a non-transferable, single-use on-chain asset. It carries

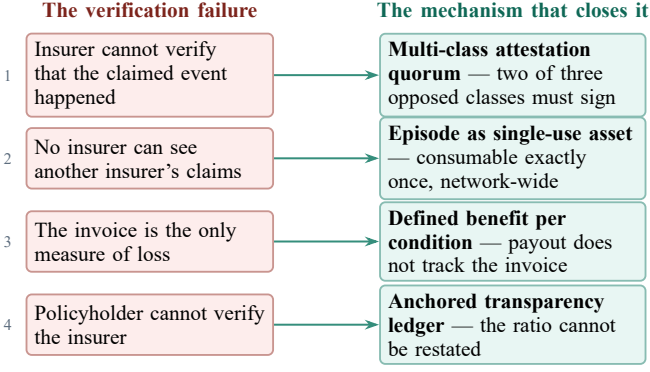


Fig. 6. The organising claim: insurance fails here for four separable reasons, and each has one mechanism assigned to it. Duplicate prevention (row 2) is one of the four, not the thesis.

Provider accreditation and independent appeals (Mechanisms 3 and 5) harden this set rather than closing a distinct failure of their own.

a commitment to the insured party's identity, a time window, a covered-category code, attestations from distinct stakeholder classes, and a final settlement state. It opens when somebody asserts the event, is attested by parties with opposed interests, and is *consumed* on settlement.

Only three parameters change between lines of business. Nothing else in the protocol changes at all:

- **The uniqueness key** — what makes two assertions the same event.
- **The attesting classes** — who can corroborate, and which of them must not be the payee.
- **The settlement basis** — what sets the amount, given that it must not be a document the claimant wrote.

Table II gives the three parameters for four lines. That is the whole generalisation claim: supply the three, and the line runs on the protocol unchanged.

Two points of honesty about scope. First, the key is strongest in health, where a person can only be in one place, so a second assertion is refused at commit rather than caught after payment. It is weakest in property, where one storm can legitimately damage several insured assets. We do not claim equal fraud resistance across lines, and Section XII says where it is weaker. Second, crop is the line where the protocol is *most* complete, not least. A parametric trigger removes claimant-supplied evidence altogether, so Failure 3 disappears rather than being reduced.

C. Two objects, not one: the event and the entitlement

The uniqueness claim is easy to overstate, and overstating it makes the protocol wrong in two directions at once. Obhoy therefore carries *two* objects (Fig. 7).

An **event** is the occurrence: this subject, this window. It is unique by construction, because the invariant refuses

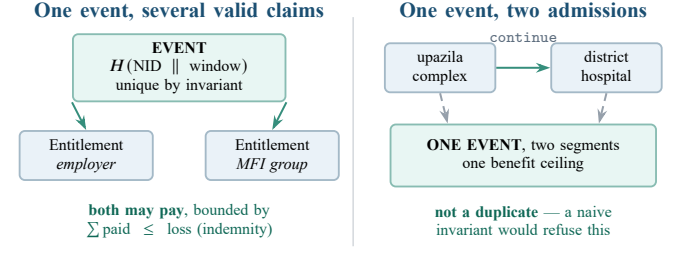


Fig. 7. Event and entitlement are different objects. **Left:** one event can carry several entitlements, one per policy, so genuine dual cover still pays; what is capped is the total recovered, not the number of claims. **Right:** an inter-facility transfer is one event with two admission segments, not two colliding events. Conflating the two objects is what makes a uniqueness invariant reject legitimate care.

a second open event on the same key. An **entitlement** is a claim of cover against that event under *one* policy. At most one entitlement exists per (event, policy) pair, and settlement consumes it.

The relation is many-to-one, and it has to be. **Several entitlements against one event are often legitimate.**

A garment worker covered by an employer scheme and enrolled through an MFI group holds two policies. A fixed-benefit hospital-cash product is designed to pay alongside an indemnity policy. Paying both is the contract, not a fraud. A protocol that blocked the second payment because the event was “already claimed” would deny valid cover, and a regulator would reject it on first reading.

What is fraudulent is *indemnity duplication*: recovering the same economic loss twice. That is an arithmetic condition over the entitlement set, not a uniqueness condition over events, and it is enforced as one:

$$\sum_{c \in I(e)} \text{paid}(c) \leq \text{loss}(e), \quad \sum_{c \in \mathcal{F}(e)} \text{paid}(c) \leq \text{cap}(e) \quad (1)$$

Here $I(e)$ and $\mathcal{F}(e)$ are the indemnity and fixed-benefit entitlements on event e , $\text{loss}(e)$ is the assessed loss, and $\text{cap}(e)$ is an aggregate declared in advance. This is coordination of benefits, and it is what the mechanism really does: **the ledger does not forbid the second claim, it lets the second insurer see the first** — which today no insurer can, at any price.

The same split repairs a false positive that would otherwise break the health deployment outright. A patient stabilised at an upazila health complex and then moved to a district hospital produces two admissions and one clinical episode. A naive invariant would see the receiving facility's assertion collide with the open event and refuse it, denying a valid claim at the worst possible moment. The event therefore allows a `continueEvent` transition. The receiving provider, attested by the trans-

TABLE II
ONE PRIMITIVE, FOUR SETTINGS. ONLY THE THREE PARAMETERS BELOW CHANGE BETWEEN LINES. THE CHAINCODE, GOVERNANCE MODEL AND TRANSPARENCY LAYER DO NOT.

Line	Uniqueness key	Attesting classes (two of three, one non-payee)	Settlement basis
Health	$H(\text{NID commitment} \parallel \text{admission window})$ — a person cannot be admitted twice at once	Admitting hospital; independent clinician or diagnostic centre; MFI/NGO field agent at the bedside	Defined benefit per diagnosis—procedure group, published in advance
Crop	$H(\text{parcel ID} \parallel \text{season})$ — one parcel yields once per season	Automated weather station; agricultural extension officer; MFI field agent or satellite index provider	Parametric: index breach at the registered station, no claim document at all
Motor	$H(\text{vehicle ID} \parallel \text{incident timestamp})$ — one collision per vehicle per moment	Police or traffic authority report; independent surveyor; workshop (payee, cannot attest alone)	Scheduled benefit by damage class, or parametric total-loss trigger
Property	$H(\text{parcel or asset ID} \parallel \text{peril window})$	Fire service or disaster authority; independent loss adjuster; local government record	Parametric for named catastrophe perils; scheduled otherwise

ferring one, adds an admission segment instead of opening a new event. A readmission inside a set window links the same way and is judged against a single benefit ceiling. **The invariant blocks duplicate payment, not repeated contact with the health system.** Getting that boundary wrong is how systems of this kind fail in the field.

D. Obhoy in outline

Obhoy has two layers, and mixing them up is the most common way this kind of proposal is misread. **Layer 1 — the products** the infrastructure makes possible: low-premium, group-enrolled cover — defined-benefit catastrophic hospitalisation, parametric crop, scheduled motor — each an instantiation of the same primitive, of which hospitalisation is the one implemented first. **Layer 2 — the infrastructure:** a permissioned network on which any licensed insurer, mutual, takaful operator or government scheme can settle claims against one shared record. **Obhoy is infrastructure, not an insurer.**

E. Design constraints, and why each earns its place

Every constraint below removes an attack surface instead of policing it.

Discrete, high-value, physically checkable events only. Outpatient visits, minor dents and small spoilage claims are frequent, cheap and hard to check — that combination is how community schemes fail worldwide. The protocol targets the rare, expensive, checkable event that causes financial ruin.

Settlement independent of claimant-supplied evidence. Parametric where an index exists, defined-benefit otherwise. If the payout does not depend on a document the claimant wrote, **inflating that document earns nothing.** Failure 3 closes *on the amount* by construction. The remaining attack moves to the category code, which Mechanism 2 and outlier detection reduce but do not remove.

Group enrolment only, through MFIs, NGOs and employers. This blocks adverse selection and rides a channel that already reaches the target households. The 2014 crop pilot showed the same channel can carry premium and payout through bKash [10].

Risk carried by licensed insurers, or a member-owned mutual. Obhoy carries no insurance risk. A second configuration supports **takaful** mutuals, whose credibility rests on transparent accounting and which therefore gain most from a verifiable ledger.

F. The five mechanisms

Mechanism 1 — the event as a single-use asset. On assertion, an event opens under one global invariant: *no two open events may share a uniqueness key*. In health that mirrors physical reality, since a person cannot be admitted to two hospitals at once. The invariant runs on the key of Table II and never on the category code, so changing the claimed condition does not evade it. Settlement consumes the *entitlement*, and the bound in (1) governs what any further entitlement on the same event may pay.

Mechanism 2 — multi-class attestation quorum. Nobody is paid on the payee’s word. Settlement needs attestation from **at least two of three stakeholder classes**, and at least one of those must not be getting paid (Table II). The quorum lives in chaincode and in the endorsement policy, not in a policy document.

Mechanism 3 — provider accreditation as a revocable credential. A hospital, workshop or assessor with a pattern of failed attestations is flagged and can be de-accredited on-chain, permanently and visibly, unable to re-register clean to shed the record.

Mechanism 4 — published reserves and settlement ratio. Each pool publishes, per period, written premium, claims received/settled/denied with reasons, mean settlement time and reserve position, anchored outside the consortium’s control (Section V-E). **This attacks Failure 4 directly, and no conventional Bangladeshi insurer offers it today.**

Mechanism 5 — independent appeals. Denial stays a human judgement; no ledger can force an insurer to accept that an event qualifies. Every denial is recorded with a coded reason and sent to an independent panel whose membership and decisions are themselves on-chain — a real limit of the technology, bounded rather than hidden.

G. Fraud taxonomy, and the assumptions it rests on

Table III maps each known attack to the control that addresses it. **Every guarantee there is conditional, and on four assumptions.** We state them once so each row can be read against them:

- A1 Membership.** The asserting provider and the paying insurer are both network members. A claim settled entirely off-network is invisible to the ledger, and no uniqueness guarantee reaches it.
- A2 Identity resolution.** One person resolves to one subject commitment everywhere. Duplicate or forged national identities defeat the invariant *before* chaincode sees it, and chaincode cannot detect them.
- A3 Attestation independence.** At least one attesting class does not depend economically on the payee.
- A4 Key custody.** Credentials sit with the party they name. A stolen custodial key forges attestations of the class it holds.

Where the table says no residual risk, it means **none under the stated assumption**, not zero risk in the world. The assumptions are the attack surface. **A2 carries the strongest guarantee in this paper:** the protocol turns a claims-integrity problem into an identity-integrity one.

TABLE III
FRAUD TAXONOMY: ATTACK, CONTROL, RESIDUAL RISK. “NONE” MEANS NONE UNDER THE STATED ASSUMPTION (SECTION IV-G), NOT ZERO RISK IN THE WORLD.

Attack	Control → residual risk
Same event at two insurers	Uniqueness invariant → <i>none under A1–A2; residual is identity aliasing</i>
Same event twice, one insurer	Entitlement consumed on settlement → <i>none under A2</i>
Duplicate recovery under genuine dual cover	Coordination bound (1) → <i>none under A1</i>
Inflated invoice or repair quote	Parametric or defined benefit → <i>none for amount; category remains attackable</i>
Upcoding to a higher-paying category	Independent-class attestation + outlier detection → <i>partial, audit needed</i>
Fabricated event (ghost patient, phantom loss)	Two-of-three, non-payee required → <i>low under A3–A4</i>
Payee-verifier collusion	Rotation, anomaly detection, audit sampling → the real residual risk
Ghost policy / enrolment fraud	Non-transferable, NID-bound credential → <i>low</i>
Enrolment after the loss	Group-only + waiting period → <i>low</i>
De-accredited party re-registers	Permanent on-chain history → <i>low</i>
Insurer over-denies	Published statistics + independent appeals → <i>partial</i>
Insurer restates its settlement ratio	Public Merkle anchoring → <i>none for anchored periods</i>
Consortium capture	Weight caps, NC = 3 (Section VI-B) → <i>low</i>

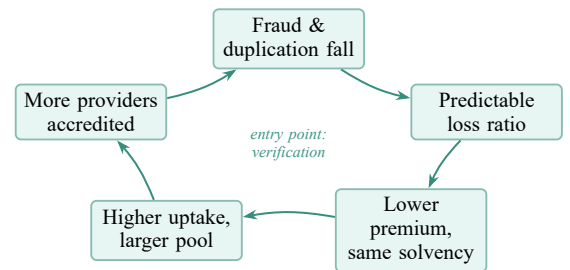


Fig. 8. The positive reinforcing loop. Its entry point is verification, not capital — which is why a student-scale pilot can start it turning.

That is a real reduction, not a dissolution. Two further rows are not claimed as solved at all.

H. Differentiating factor: the reinforcing loop

The mechanisms compose into a loop no conventional insurer can copy (Fig. 8). Fraud and duplication fall, so the loss ratio becomes predictable. Predictable losses allow a lower premium at the same solvency margin. A lower premium plus a *checkable* settlement record raises uptake among buyers who assumed insurers never pay. Higher uptake enlarges the pool, which cuts variance and unit administrative cost. Larger, cheaper pools pull more providers into seeking accreditation, which widens attestation coverage and cuts fraud again. The loop starts at verification, not at capital. That is why a student-scale pilot can enter it and not only a large incumbent.

V. ARCHITECTURE

A. Platform selection and rationale

Obhoy runs on **Hyperledger Fabric**, with periodic public anchoring. Four arguments support that choice.

Why permissioned. Every participant is a known, licensed, legally accountable body: insurers licensed by IDRA, hospitals registered with the DGHS, MFIs regulated by the MRA. Nobody needs anonymity; everybody must be identifiable to the regulator. A public permissionless chain cannot offer that together with the confidentiality health data demands.

Why Fabric. Three Fabric features match requirements that would otherwise need custom engineering. **Private data collections** [18] let competing insurers work on one channel while keeping commercial terms hidden from rivals. **Idemix** credentials [19] let a field verifier or policyholder submit an attestation anonymously within their own MSP, proving class membership without revealing which individual acted. Anonymity stops at the client: Fabric does not extend Idemix to chaincode endorsement, so Section V-B's quorum is still enforced by ordinary X.509 peer identities, and Idemix's own custom-attribute and revocation support remain limited [19]. **Pluggable endorsement policies** [20] express the multi-class quorum of Section IV-F as a protocol rule, not as application code that could be bypassed.

Why also anchor publicly. A consortium can be captured or legally compelled. If the transparency totals live only inside it, it can rewrite them, and the whole trust proposition collapses. Obhoy therefore commits a Merkle root of each period's totals to a public chain (Polygon, or Bitcoin via OpenTimestamps [21]). **Confidentiality inside, immutability outside. Neither half works alone.**

Why not a conventional database. Answered against the strongest alternative in Section III-C.

B. Network topology and consensus

The institutional layout is drawn in Fig. 11's blockchain layer; this subsection states the rules behind it.

Ordering service. Raft-based [22], with five orderer nodes spread across different institutional classes, so no one class holds an ordering majority: O1 IDRA, O2–O3 participating insurers, O4 the MFI/NGO aggregator consortium, and O5 an independent academic node at a public university. Raft is crash-fault tolerant, not Byzantine. Institutional spread bounds who can stall the service; it does not make a malicious ordering majority harmless.

Peers. Every insurer, the provider association, each aggregator, IDRA and the academic auditor run endorsing peers. Providers connect through the association's peer or a hosted gateway. Making every upazila hospital run its own peer would block adoption and add no security.

Channels. One main channel carries policy credentials, the event registry and settlement state, with private data collections separating per-insurer detail inside it. A separate audit channel carries only the transparency totals, so the regulator and the academic auditor read continuously without seeing commercial detail.

Endorsement policy for settlement (formula in Fig. 11): the `OutOf(2,...)` term is the two-of-three quorum at protocol level; `InsurerMSP` signs as payer, not as a third attesting class. The non-payee rule is runtime-dependent, so `closeEvent` enforces that half in chaincode (Appendix A).

C. Asset tokenisation

Obhoy tokenises **the insurable event itself**, and separately the entitlement claimed against it. This is the central design decision. It is what makes property P2 of Fig. 5 possible at all: refusal at commit rather than detection afterwards.

An event is a *non-transferable, single-use* on-chain asset whose deterministic key comes from the domain's uniqueness parameter (Table II). In the healthcare pilot:

$$\text{eventKey} = H(\text{NIDCommitment} \parallel \text{admissionWindow}) \quad (2)$$

NIDCommitment is not a salt chosen by each insurer. Independent salts would mint a different commitment per insurer for the same person, silently breaking cross-insurer Mechanism 1; one global salt would instead make

NIDs guessable and every commitment linkable. Obhoy computes it as a keyed PRF, $\text{HMAC}_{K_v}(\text{NID} \parallel \text{context})$, with K_v HSM-held under threshold custody split across institutional classes, so no single class computes a commitment alone. context domain-separates each use (event key vs. policy credential), so a leak in one does not unlock the other. Key version v travels with the commitment; a suspected compromise retires K_v and forces re-commitment under K_{v+1} through the enrolment path already used for custodial-key recovery (Section V-J).

The key **leaves out diagnosis and provider** on purpose, so changing either cannot mint a “different” event. Four properties follow, each enforced in chaincode rather than by convention:

- **Uniqueness (minting).** `openEvent` fails if an unconsumed event already exists for that key. Minting is limited by physical reality, not by policy — subject to A2, since two identities for one person defeat the check before chaincode sees it.
- **Continuation, not duplication.** `continueEvent` adds an admission segment to an open event when a patient is transferred or readmitted inside the set window, on attestation by the transferring provider. Without it the invariant would reject legitimate care (Section IV-C).
- **Non-transferability.** An event is bound to the subject commitment that authorised it. There is no transfer function, so it cannot be sold, assigned or repointed to another policyholder.
- **Consumption (burning).** Settlement consumes an *entitlement*, one per (event, policy) pair. Nothing re-opens a consumed entitlement, and any further entitlement on the same event is bounded by (1). The same loss cannot be recovered twice, and genuine dual cover still pays.

Two more asset classes follow the same rules: the **policy credential** (non-transferable, bound to an NID commitment, revoked on lapse) and the **provider accreditation credential** (revocable, with a history that survives revocation, so a de-accredited facility cannot re-enter clean). Obhoy issues **no fungible or tradable token**. There is no cryptocurrency and no speculative instrument. All money moves in taka through regulated mobile financial services and banking rails, outside the ledger.

D. On-chain and off-chain data model

Table IV governs every write. **No single on-chain field is protected health information on its own.** A diagnosis appears only as a membership proof against

TABLE IV
ON-CHAIN VERSUS OFF-CHAIN DATA PLACEMENT

On-chain	Off-chain
Policy credential (NID commitment)	NID numbers, names, addresses
Event record, state, timestamps	Clinical notes, imaging, labs
Attestation signatures + class	Itemised invoices
Benefit schedule version hash	Insurer actuarial models
Settlement authorisation, denial code	Bank / MFS account details
Provider accreditation history	Biometric templates
Period transparency aggregates	Documents (hash on-chain only)

a coded benefit group; identity only as the keyed-PRF commitment of Section V-C, per (2). **That is not the same claim as anonymity.** CategoryCode, SubjectCommitment, provider identity and timestamps together are a metadata surface: a party seeing enough of them, over enough periods, can plausibly link a subject to an illness category without reading a diagnosis field. Hashes and pseudonyms are not automatically anonymous. The mitigation is architectural, not a claim of zero risk: private data collections scope who sees which combination (Section V-B), and Section V-K treats the residual as a DPIA question. Off-chain data sits in an encrypted store held by the controlling institution, with only a content hash and an access envelope on-chain.

E. Public anchoring

At the close of each settlement period the network builds a Merkle tree over that period’s transparency totals and commits the root to a public chain. Anyone — journalist, regulator, prospective policyholder, donor — can check a published claims-paid ratio against what was committed at the time, and confirm that no earlier period has been rewritten. One root per period costs almost nothing, and the technique already works at far larger scale than Obhoy needs [21]. **Anchoring binds integrity after inclusion, not completeness.** A claim that never reached the ledger is absent from the tree, and the root commits faithfully to a record missing it. A1 narrows that gap rather than the anchor: writes come from parties with opposed interests, so omission needs all of them to agree to it.

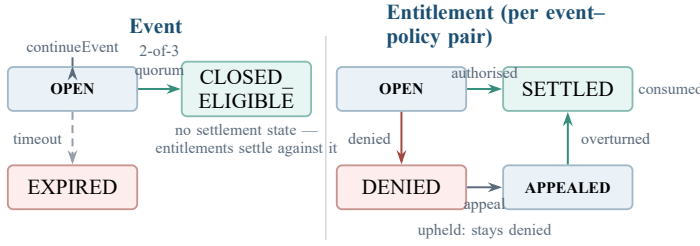


Fig. 9. Two objects, two machines. The **event** (left) has no settlement state: it becomes eligible once and stays eligible, so several entitlements can settle against it in turn (Fig. 7). The **entitlement** (right) is what settlement actually consumes, one per (event, policy) pair — the burn property of Section V-C.

F. Event lifecycle and state machine

Fig. 9 draws the two objects of Section IV-C as two machines, because one machine cannot represent both correctly. The **event** reaches `CLOSED_ELIGIBLE` once and stays there — it has no settlement state of its own, so a second legitimate entitlement can still settle against it later. No transition re-opens a consumed event key: that half is the uniqueness invariant of Mechanism 1. The **entitlement** is what settlement actually consumes, one per (event, policy) pair; apart from an upheld denial, `SETTLED` is its only terminal state. That half is the burn property of Section V-C.

G. Claim lifecycle sequence

Fig. 10 traces one event end to end across all seven participant roles, so the mechanism of Section IV-F can be read at a glance.

H. Smart contract (chaincode) architecture

Chaincode is written in **Go** across the eight contracts of Fig. 11. Entitlements and the bound in (1) live in `ClaimSettlement`; events and their continuation segments live in `EventRegistry` — split deliberately, since the two objects have different uniqueness rules.

`EventRegistry` carries the invariant: `openEvent` checks the policy is active and the asserting party accredited, then that no unconsumed event already exists for that key, before creating the event in state `OPEN`. `attestEvent` verifies the attester’s credential and class and rejects a class already present, so a payee cannot fill the quorum alone. `closeEvent` refuses to advance unless the attestations span two classes including one non-payee. Appendix A states these formally as the prototype’s negative-test suite.

On-chain event state: `EventID`, `SubjectCommitment`, `PolicyID`, `AsserterID`, `CategoryCode`, `OpenTS`, `CloseTS`, `State`

$\in \{OPEN, CLOSED_ELIGIBLE, SETTLED, DENIED, APPEALED, EXPIRED\}, Attestations[], ScheduleVersionHash, SettlementRef.$

I. Layered system architecture

Fig. 11 is the whole pipeline in one figure: five layers between the actors and the external rails, with the claim path traced through all of them. Nothing in it is new; it composes what Sections V-B through V-C already established.

J. Privacy, key management and access control

Identity privacy. A policyholder appears on-chain as the keyed-PRF commitment of Section V-C, not a per-insurer salt. Only the enrolling insurer and aggregator hold the mapping back to a real identity, inside a private data collection; the PRF key itself is held separately, under threshold custody, by neither of them alone.

Data privacy. Diagnoses are never written in the clear. In the MVP the adjudicating insurer sees the diagnosis group through selective disclosure of a verifiable credential. The roadmap target is a zero-knowledge proof that the diagnosis falls in the covered set **without revealing which condition it is**. That is roadmap, and we do not claim it in the prototype.

Key management — the honest problem. This population cannot be asked to manage private keys. A scheme that costs a widow her cover because she lost a seed phrase is worse than no scheme. **Policyholders** therefore use custodial keys held by the aggregator, recovered through the MFI field agent plus a second household member — the social recovery MFS users already know, and why A4 is an assumption rather than a guarantee. **Providers and verifiers** use device-bound keys in mobile secure elements with on-chain revocation. **Institutional members** use HSMs with documented rotation [23].

Access control. Role-based, enforced by Fabric MSP and channel policy [20]. Field verifiers attest but never see clinical data. Providers open and attest but cannot authorise payment. Insurers adjudicate only their own policies. IDRA reads across the audit channel. The academic node reads totals only.

K. Regulatory compliance in a regulated industry

Obhoy is **not** a risk carrier and needs no insurance licence. Licensed insurers underwrite; Obhoy is claims infrastructure. IDRA joins as a network member with continuous read access instead of an outside inspector receiving periodic reports, so supervision becomes real-time. Solvency and reserve reporting comes from ledger

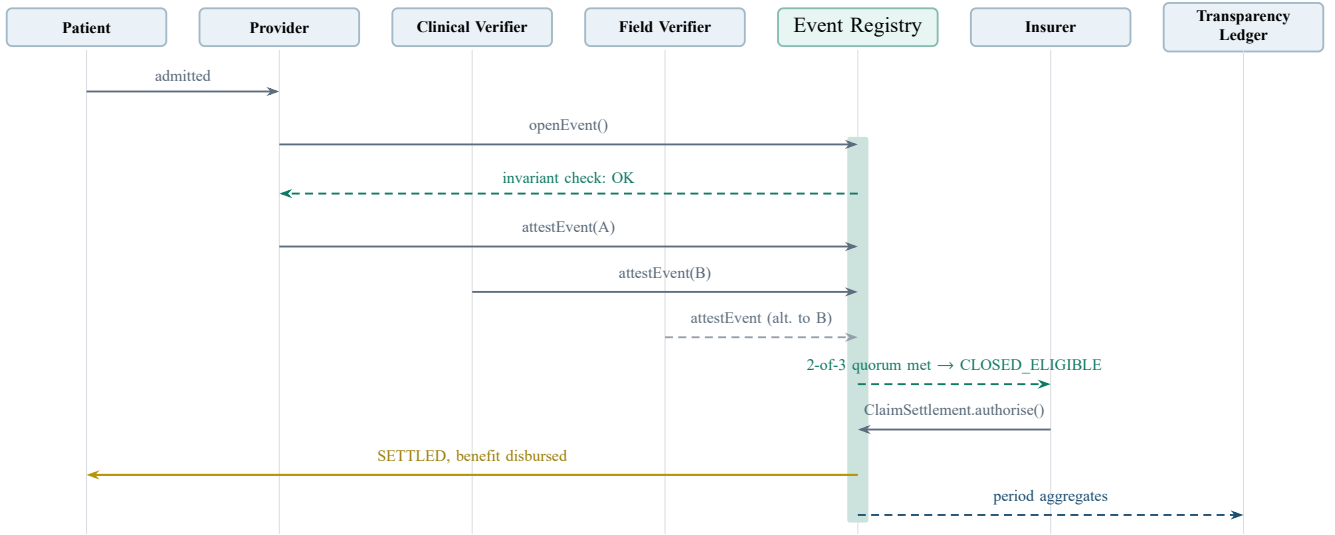


Fig. 10. Claim lifecycle: admission → attestation → quorum → settlement → transparency publication. Quorum is **two of the three** attesting classes, at least one non-payee; Field Verifier’s dashed arrow shows it as an alternate second signer, not a required third one.

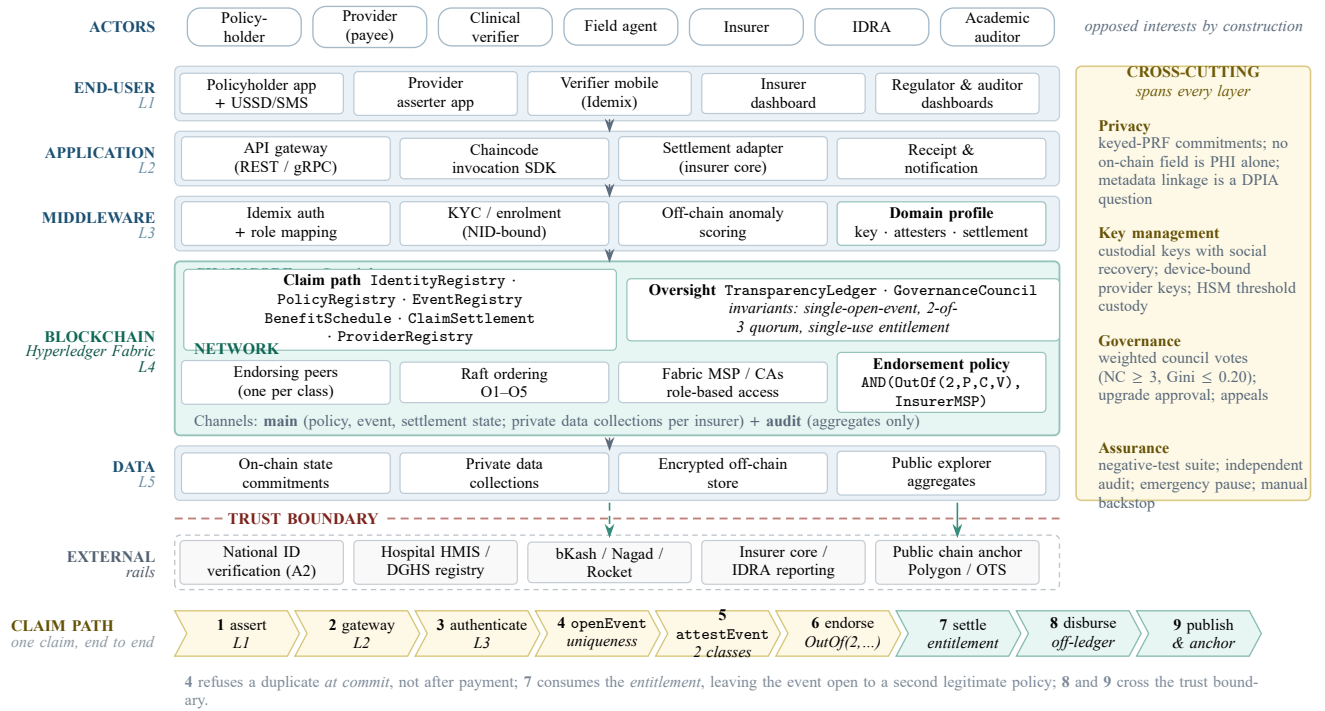


Fig. 11. Obhoy end to end. Five layers sit between the actors above and the external rails below; the ribbon traces a single claim through them, with the invariants that matter noted inline. The dashed rule is the **trust boundary**: everything beneath it is a system Obhoy integrates with but does not control, which is where assumptions A1 and A2 come from (Section IV-G). The domain profile is configuration, not a separate service — the three parameters of Table II are all that change between lines of business.

state rather than self-declaration, which removes a whole class of reporting error.

Bangladesh’s Personal Data Protection Act, 2026 [24] replaced the interim Ordinance of the previous November [25]. It treats health data as sensitive personal data

under stricter consent and processing rules, and sets up a National Data Management Authority over data fiduciaries. Obhoy’s field-level design is built toward it — no on-chain field of Table IV is itself protected health information — but that is a design intent, not

a compliance finding, given the metadata-linkage risk noted in Section V-D. Court-ordered disclosure runs against the off-chain stores under defined process, and the ledger records only *that* a disclosure happened. **This is an engineering posture, not a legal opinion.** A documented legal basis per field, a retention/deletion schedule, and a formal DPIA are Phase 0 gates alongside counsel review, not a later formality.

L. Legacy and API integration

Obhoy is API-first and attaches to existing systems rather than replacing them: hospital HMIS and DGHS facility registries for admission and discharge events; the national ID verification service for enrolment, which is also where A2 is discharged; **bKash / Nagad / Rocket** for premium collection and payout; MFI core banking for group enrolment; insurer policy administration through a settlement adapter; IDRA supervisory reporting; and SMS/USSD fallback for policyholders without a smart-phone.

Payment-rail safety. Disbursement crosses a boundary the ledger does not control, so every instruction carries a unique payload-bound request identifier: the MFS adapter rejects a repeat request with a different payload and returns the original receipt for a valid retry. A disbursement with an unknown outcome stays pending, reconciled against the daily settlement file rather than silently retried. **The ledger authorises payment; it does not execute it.**

VI. GOVERNANCE

Governance is part of the architecture here, not a policy annex, covering network membership, business governance, technology governance and interoperability in turn.

A. Network membership governance

Member on-boarding. Insurers join on proof of an IDRA licence and acceptance of the charter. Providers are accredited on DGHS registration plus an on-site assessment by the provider association and one independent member. Field verifiers are credentialed by their MFI or NGO after standard training, and enrol policyholders through group channels with national ID verification.

Member off-boarding. A two-thirds council vote can suspend any member; loss of the underlying licence suspends one automatically, as does hitting set thresholds of failed attestation or upheld appeals for providers. Every event is a permanent ledger record: suspension revokes

TABLE V
GOVERNANCE VOTING WEIGHTS BY STAKEHOLDER CLASS

Stakeholder class	Weight
Insurers (collectively)	0.30
IDRA (regulator)	0.20
MFI / NGO aggregators	0.20
Provider association	0.15
Independent academic auditor	0.15

the MSP credential without deleting past signatures, and a de-accredited facility cannot re-enter clean.

Regulatory oversight. IDRA is a first-class network member. It runs orderer O1 and an endorsing peer, and reads continuously across the audit channel. Oversight is architectural rather than contractual: the regulator does not request data, it already holds a validating copy.

Permission structure. Role-based, enforced by Fabric MSP and channel policy rather than application code, so no class can assert a fact outside its real-world role (Section V-J).

Network operations. Peers and orderers sit across five institutional classes (Section V-B), and no member runs a majority of either. The charter fixes node service levels, security baselines, certificate authorities and incident-response duties. Compliance is reviewed each period alongside the transparency totals.

B. Governance voting weights and measured decentralisation

These weights govern the GovernanceCouncil chaincode's own votes — charter amendments, upgrade approval, member admission and off-boarding — **not Fabric transaction validation.** Block validation runs on the OutOf(2,...) endorsement policy and Raft ordering of Section V-B, which no weight table alters. Power over the council's votes is spread across the five classes of Table V under a hard cap: **no class may exceed 0.30**. Sorted cumulative weight runs $0.30 \rightarrow 0.50 \rightarrow 0.70$. Two classes cannot reach a majority and three can, so the **Nakamoto Coefficient is 3**. The Gini coefficient of the same weights, $G = \sum_i \sum_j |w_i - w_j| / 2n^2 \bar{w} = 1.40/10$, is **0.14**, where 0 is perfect equality (Fig. 12). Both beat a published three-class consortium model reporting weights 0.4/0.3/0.3, Gini 0.26 and Nakamoto Coefficient 2 [26]; the method follows established practice [27].

Both metrics are recomputed and published every period alongside the transparency totals, and they bind rather than describe. A new member class is admitted only if admission keeps the Nakamoto Coefficient at 3

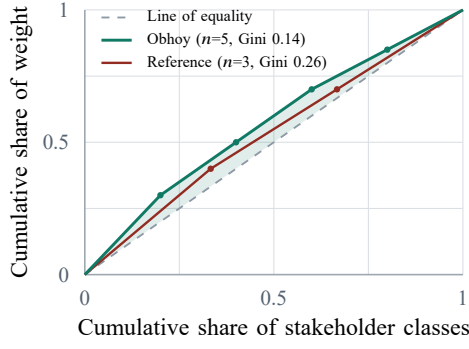


Fig. 12. Lorenz curve of governance voting weight. The shaded wedge is what the Gini coefficient measures — a flatter curve sits closer to the diagonal and concentrates less governance power.

Reference series: a published three-class consortium design [26].

Method follows [27].

or above and the Gini at or below 0.20. Both measure concentration of *formal* authority, not independence of incentives: classes can be constitutionally separate and still align commercially. A floor on dispersion, not proof against capture.

Conflict rules, both in chaincode rather than in policy: the provider association may not help validate claim decisions affecting its own member facilities, and insurers may not help validate appeals against their own denials.

C. Business network governance

Charter and operating structure. Obhoy runs under a written charter administered by the Obhoy Governance Council. Seats go by stakeholder class, and a rotating chair cannot be held by the same class in consecutive terms. IDRA acts as regulator. Licensed insurers and mutuals carry risk. Obhoy is neutral infrastructure and is barred from underwriting.

Common services. The council owns four shared services: the versioned benefit schedule, so repricing cannot run backwards; the appeals panel pool; the transparency-aggregation and anchoring service; and the off-chain fraud-scoring engine, disclosed to members and auditable by the regulator.

Service levels and appeals. Settlement must happen within a set number of days of event closure; a breach is recorded on-chain and shows in the insurer's public record the same period, not at the next annual filing, since performance comes from ledger state rather than self-reporting. A denied claim can go to a panel drawn by lot from independent clinicians and consumer representatives, excluding anyone tied to the denying insurer; an insurer that over-denies is visibly reversed.

Sustainability. The platform fee (Section XI) falls on insurers, never on policyholders, so platform economics cannot cut a household's benefit.

D. Technology infrastructure governance

Distributed IT structure. Orderers and peers sit across institutional classes (Section V-B), so no single member can halt the network alone; Raft gives crash-fault tolerance, not protection from a malicious majority, so certificate and upgrade governance are specified here rather than assumed.

Technology adoption. Chaincode upgrades need council approval and an independent security audit, published alongside the proposal. A high-risk upgrade touching settlement or invariant logic needs concurrent approval from IDRA, an insurer and a non-financial class. An emergency pause can stop new events while letting open ones settle or appeal through a documented recovery path.

Data services. Governed by the split in Table IV, amendable only by supermajority. Private data collections copy values to authorised peers only, but a compromised peer can still read what it is entitled to; sensitive evidence is therefore encrypted at the application layer, with only a content hash and access envelope on Fabric.

Risk mitigation. Third-party contract audit before any production deployment; sandbox certification for every partner integration; continuous monitoring with defined incident response; and a documented manual backstop, so a chaincode defect freezes state rather than mis-settling it.

E. Governance of interoperability

Partners work to standard API contracts and binding data-sharing agreements that set purpose limitation and retention in line with Section V. Sandbox certification against a published conformance suite is mandatory before any production integration, and all API connections are monitored with alerting. A new partner category is admitted only if it keeps the Nakamoto Coefficient at 3 or above and the Gini at or below 0.20.

VII. MARKET

A. Market definition

In the initial pilot domain, Obhoy addresses three connected markets.

Unprotected health spending. Households spent an estimated Tk 34,400 crore on drugs alone in the year covered by the 6th BNHA, plus Tk 7,200 crore on

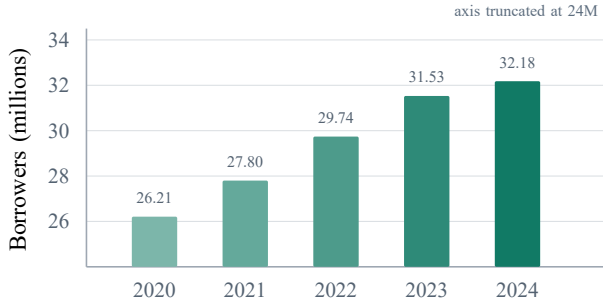


Fig. 13. Growth of the MFI borrower base — the natural group-enrolment channel, and already the largest verified financial relationship network reaching the target population. The vertical axis begins at 24 million.

Source: IMF Financial Access Survey / Microcredit Regulatory Authority [31].

ambulatory care [7]. That is not addressable revenue; it is the size of the risk nobody insures.

The insurance market that does not yet exist. Penetration is 0.33% of GDP [1]. The market is defined by its absence, so the opportunity is largely *created* rather than captured. That is a stronger position than fighting for share of an existing pool.

Government health-protection programmes. SSK and any successor need exactly the claims verification Obhoy provides [28]. Its own independent evaluation names administrative delay in claim submission and reimbursement as the binding constraint on use [29]. That is a settlement-layer problem, not a coverage-design one. Public schemes are the largest single contract available.

B. Market size and serviceable segments

Total insurance premium across all lines was Tk 18,534 crore in FY2024–25 [1], and the SSK pilot reaches about 100,000 households in three upazilas [28]. Set against the household spending above, that is the gap between exposure and cover. The serviceable channels are the RMG workforce of some 4.4 million [30] and, above all, microfinance.

The MFI channel is the critical number (Fig. 13). Bangladesh’s microfinance sector already holds verified, grouped, regularly-contacted relationships with 32.18 million borrowers [31]. Those are the households most exposed to catastrophic health costs. Obhoy rides that channel instead of building one.

C. Domain coverage and geographic reach

The protocol reaches a new setting along two axes, and they carry different risk. A new *domain* reuses the same consortium, governance and chaincode, and

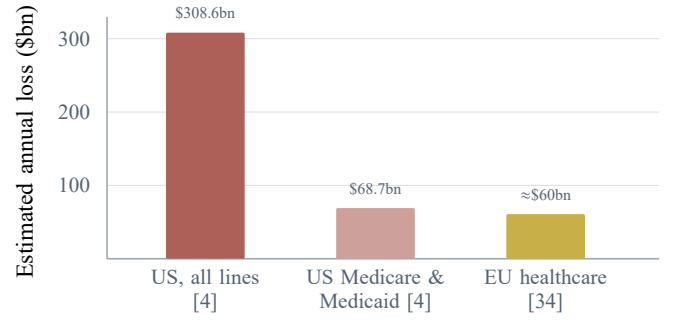


Fig. 14. Fraud loss in mature verification regimes, redrawn from the cited sources. These jurisdictions do not lack technology; they price the loss into the premium instead. Bangladesh cannot, because the loading exceeds what the customer can pay.

Note. Bars are estimates from different bodies on different methodologies and are not strictly comparable; they are placed on one scale to show order of magnitude only. EHFCN’s EUR 56bn is shown converted at approximately 1.08 USD/EUR. India’s PM-JAY is reported in cases rather than currency [32], [33] and is therefore not plotted.

changes only the three parameters of Table II. A new *country* reuses the architecture but rebuilds the consortium from nothing. Domains are the cheaper axis.

Crop shows how little has to change between domains. The payment rail is proven: the 2014 weather-index pilot paid claims through bKash [10], the same channel Obhoy uses. Part of the attesting infrastructure exists too: that pilot’s twenty automated weather stations produce exactly the kind of evidence Failure 3 needs, since no claimant writes them, and parametric settlement removes the inflation attack rather than reducing it. What the pilot lacked was trust — farmers called it a “Ponzi scheme” [2] — and that is the gap Mechanism 4 attacks.

D. International scalability

The failure is not specific to Bangladesh: every emerging universal-health-cover programme meets claims fraud as its main operational problem (Fig. 14). India’s Ayushman Bharat had detected 342,988 fraud cases and de-empanelled 2,359 hospitals at its most recent disclosed figures [32], [33], catching fraud *after* payment because its architecture has no structural duplicate prevention. The EHFCN puts EU healthcare fraud and corruption near EUR 56 billion a year [34]. The architecture transfers directly to Indonesia, Nigeria, Pakistan, the Philippines and Kenya.

VIII. PARTNERS AND STAKEHOLDERS

No organisation named below is a confirmed partner. Each row names a partner class, what it must contribute,

TABLE VI
STAKEHOLDERS AND PARTNERS: PAIN POINT TODAY, AND ROLE IN THE NETWORK

Party	Pain point today	Strategic role
Low-income households	Pay cash; sell assets or borrow; forgo treatment	—
IDRA	Supervises on self-reported, annual, unverifiable data	Regulator and network member; real-time oversight
Licensed insurers	Cannot verify claims; must load premiums for fraud	Underwrite risk; the platform's paying customers
Hospitals & provider association	Cash-paying patients, high default, delayed payment	Attestation; volume incentive drives accreditation
MoHFW / HEU	Public schemes face the same verification problem	Policy owner; largest potential scheme partner
MFIs and NGOs (BRAC, Gonoshasthaya Kendra)	Health shocks push client households into default	Group enrolment, field verification, client trust
bKash / Nagad	—	Premium collection, benefit disbursement
Public university	—	Independent academic node, actuarial validation
Development partners	—	Pilot funding and international credibility
Fraudulent providers, claim syndicates (adversaries)	Depend on today's inability to verify or cross-check	—

and the incentive that Phase 0 of Section X has to validate.

A. Stakeholders and strategic partners

Health shocks are a documented cause of microfinance distress. Sickness was the stated reason for difficulty among 18.75% of currently-due borrowers, and about one-third of late BRAC instalments are attributed to family health emergencies [35]. **MFIs are not just a distribution channel. They have a direct portfolio interest in this product working. Gonoshasthaya Kendra** is the closest domestic precedent: a community health-insurance scheme since 1973, with contributions scaled to ability to pay [36], so its model is single-

provider and paper-based, lacking the portability and cross-checking Obhoy adds.

B. Partner on-boarding strategy

The order is deliberately lopsided. A policyholder-facing app without insurer and regulator participation would be one more promise nobody can check.

IDRA comes first, framed correctly as supervisory technology, not a fintech seeking a licence exemption. **Insurers** get a loss-ratio argument, not a technology one: current fraud loading against the platform fee, plus access to a mass market closed to them today. **MFIs and NGOs** get portfolio protection plus a service fee; **hospitals** get insured volume and faster, guaranteed settlement. A government scheme running on Obhoy should make accreditation a condition of empanelment, turning a voluntary request into a commercial necessity.

Outreach asks partners to *test assumptions*, not endorse the proposal: written technical feedback outweighs a letter of support.

IX. VALUATION

A. Value generation

The main output is prevented impoverishment. A household that would have sold its land to pay for surgery does not. That can be measured: catastrophic-expenditure rates and health-related asset loss in the pilot cohort against a matched control. It should be measured, not asserted. Financial value comes from platform fees set for self-sufficiency at moderate scale (Section XI). Those fees are the enabling mechanism, not the objective.

Value reaches each party differently. **Households:** affordable cover and the ability to check an insurer's record *before* buying. **Insurers:** verifiable loss experience, mass-market access, lower adjudication cost. **Providers:** insured volume and predictable revenue instead of uncertain cash collection. **Regulator:** real-time supervision instead of annual self-reports. **Government:** verified claims with less leakage, and a verification layer it would otherwise pay to build. **MFIs and NGOs:** portfolio protection and deeper client relationships. **Adversaries:** lose the information gap they rely on.

B. Partner incentives allocated through the platform

Incentives are paid through the platform rather than settled beside it. Every one triggers on an on-chain event, and none needs off-platform reconciliation. Field-verifier fees trigger on a recorded attestation. Provider settlement releases on quorum, not on invoice submission.

Aggregator commission accrues per on-chain enrolment. Appeals-panel honoraria release on a recorded, signed decision. **The platform fee itself is computed from ledger state**, so the operator cannot over-bill an insurer without the regulator’s peer seeing it.

C. Alignment with the UN Sustainable Development Goals

In the pilot domain, **SDG 3.8** — universal health coverage *including financial risk protection* — is the primary target and maps one-to-one onto what Obhoy does. Secondary targets follow. **SDG 1**, because catastrophic health spending is a leading cause of poverty. **SDG 10**, because the scheme reaches households formal insurance excludes. **SDG 16**, through verifiable supervision and auditable institutional behaviour. **SDG 17**, where the public-private-civil-society consortium is the delivery structure itself rather than an afterthought.

These are intended contributions. A pilot must report them against a matched control before we claim any measured SDG impact.

X. DISTRIBUTION

Phase 0 — Foundation (M1–M3). Assemble the team; secure a letter of intent from one MFI/NGO aggregator and one licensed insurer; open dialogue with IDRA; complete the actuarial baseline; obtain data-protection counsel review.

Phase 1 — Design and MVP (M3–M7). Finalise the benefit schedule; build the MVP; commission an independent chaincode security audit.

Phase 2 — Single-upazila pilot (M7–M14). One upazila, one aggregator, one insurer, three to five accredited facilities, one full annual cycle. Measure claims incidence, fraud attempts detected, settlement time, and denial/appeal rates. **A second, non-underwriting InsurerMSP identity held by IDRA or the academic node submits scripted duplicate claims against live events all year**, proving Mechanism 1 under adversarial test ahead of real second-insurer participation (Section XII).

Phase 3 — Evaluation (M14–M17). Independent impact evaluation against a matched control: catastrophic-expenditure incidence, asset depletion, treatment-seeking behaviour, *including negative findings*.

Phase 4 — District scale-up (M17–M21). A second insurer joins with real underwriting risk; Phase 2 proved the invariant, Phase 4 tests whether real premium changes insurer behaviour as Section XI predicts.

Phase 5–6 — National scheme and second domain (M20–M24 onward). Position Obhoy as the claims layer for government health protection; instantiate parametric crop cover on the existing consortium (Section VII); package the architecture for comparable markets.

A. Initial pilot domain selection

The protocol is domain-general; a pilot cannot be (Section XII-B). **First implementation priority: health-care.** It offers the strongest uniqueness key, the largest documented trust deficit, the clearest social return, and a government scheme as possible anchor client. Crop is the readiest alternative context: its payment rail and part of its attesting infrastructure already exist [10], and its settlement is parametric. Motor is the least ready, not the last in a queue: the 2018 removal of compulsory third-party cover [3] means regulation has to rebuild that market before any infrastructure can serve it.

This is a choice of validation context, not a product sequence. We would pilot crop first if a licensed insurer committed to it before health: an anchor client with real volume beats our ranking. That is the explicit lesson of B3i against Insurwave.

B. Decision gates

The project will narrow, pivot or stop if any of the following happens. Stating them in advance is what makes this proposal testable rather than promotional.

- 1) An actuary concludes the defined-benefit schedule cannot be priced at a premium the target cohort can pay while remaining solvent.
- 2) No licensed insurer and no aggregator will validate the integration assumptions, in which case Obhoy remains architecture research rather than a deployment claim.
- 3) IDRA advises that claims infrastructure of this form cannot be operated without a risk-carrier licence, which would require restructuring the operating entity.
- 4) The prototype cannot enforce the single-open-event invariant, the continuation rule or multi-class endorsement under adversarial test — or enforces them so bluntly that legitimate transfers and dual cover are refused.
- 5) Pilot data shows the fraud reduction achieved is too small to move the premium, in which case the affordability thesis — not the technology — has failed.

TABLE VII
ILLUSTRATIVE UNIT ECONOMICSⁱ

Premium	BDT 1,200 / household / year
Benefit ceiling	BDT 50,000 / household / year
Claim frequency	5% of households / year
Target claims ratio	70%
Administration	20%
Platform fee	3% of GWP + BDT 50 / claim
Surplus / reserve	7%

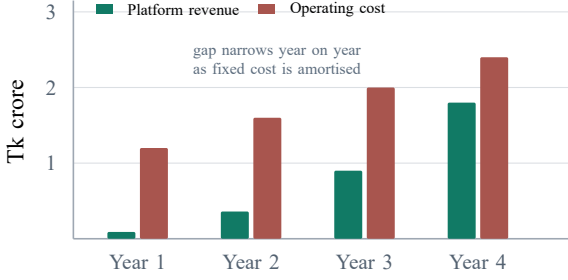


Fig. 15. Illustrative platform revenue versus operating cost, Years 1–4.ⁱ Figures are stated planning assumptions pending actuarial validation, not observed or projected traction.

XI. FINANCIALS

Every figure in this section is illustrative and markedⁱ, pending actuarial validation. They are planning assumptions that show what must be true for the model to work. They are not forecasts, and none of them is observed traction.

Revenue model. A platform fee on insurers, never on policyholders: a percentage of gross written premium plus a flat fee per settled claim. Licensing to government schemes and enrolment service fees shared with aggregators come on top. Obhoy holds no customer funds, so float is not counted as revenue.

The split is 100% only once claim frequency is priced in. At 5% frequency the flat BDT 50/claim component costs BDT 2.5 per average household-year — about 0.2% of premium — small enough to sit inside the reserve line’s own rounding. A higher claims ratio has no such slack: at 75/80/90% (5/10/20 points above target), administration and platform fee held fixed, the reserve margin runs 7% → 2% → -3% → -13% — the point past which the stop-loss layer and reinsurance of Section XII, not the pool’s own reserve, must carry the shortfall.

Stated assumptions. Households covered grow from 25,000 in Year 1 to 500,000 in Year 4; gross written premium grows correspondingly from Tk 3.0 crore to

Tk 60 crore. Fig. 15 plots the resulting revenue against operating cost.

Development-partner and government grants cover the early deficits. That is the standard financing path for health-protection infrastructure. Break-even at moderate scale works because the marginal cost of one more covered household approaches zero once the network runs. **The government scheme licence is the step change.** A national programme running on Obhoy changes these economics completely, and we present it as the upside case, not the base case.

The economic gate. An insurer joins only when the fraud loading removed is worth more than the platform fee charged. Write L for the fraud and duplication loading in the premium today, F for the Obhoy fee, and A for the adjudication cost saved by automated verification. Participation makes economic sense when

$$L + A > F. \quad (3)$$

The pilot must estimate each term from real claims data. A positive result for inpatient catastrophic cover does not carry over automatically to outpatient or low-value products. That is exactly why Section IV bounds the product scope the way it does.

XII. COMPETITION AND RISKS

A. Competition

Direct — international. Insurer-led blockchain claims consortia already exist in mature markets. Table VIII sets out each one and what it leaves unsolved. All three run on Hyperledger Fabric. **Avaneer Health** was founded by Aetna, Anthem, Cleveland Clinic, IBM and others [37]. The earlier **Synaptic Health Alliance** built a provider-directory network reporting 94% cross-member data agreement [38]. **Change Healthcare** ran a claims network in 2017–18 at up to 550 transactions per second across 900,000+ providers and 2,200 payers [39] — evidence that the platform carries this workload at national scale.

The differentiation, stated precisely. Those consortia cut administrative cost and reconcile provider directories *inside mature markets where nearly everyone is already insured*. Obhoy works where insurance barely exists, because fraud has made the premium unaffordable. Different problem, different design: defined benefit instead of reimbursement, group enrolment instead of individual, field attestation instead of EDI reconciliation, and a public claims-paid ratio used as a demand-side tool. **We are aware of no incumbent consortium that publishes member-level denial statistics.** None appears

TABLE VIII
COMPETITIVE LANDSCAPE AND RELATIONSHIP TO OBHOY

Alternative	Does not solve	Relationship
Avaneer Health	Affordability in a zero-coverage market; duplicate claiming as a market-enabling mechanism	Direct, international [37]
Synaptic Health Alliance	Claims fraud, benefit payout integrity (solves provider directory accuracy)	Direct, international [38]
Change Healthcare	Cross-payer event uniqueness; operates within near-universal coverage	Direct, international [39]
Ayushman Bharat (PM-JAY)	Structural prevention — detects fraud <i>after</i> payment	Precedent and cautionary case [32]
SSK (Bangladesh)	Scale, cross-insurer detection, public verifiability	Domestic precedent, potential customer [28]
Gono-shasthaya Kendra HI	Portability across providers, cryptographic prevention	Domestic precedent, potential partner [36]
B3i (reinsurance consortium)	Nothing — wound up 2022 for want of transaction volume	Cautionary precedent [5]
Insurwave (marine hull)	Multi-line generality; single-anchor by design	Positive precedent for launch strategy [6]
Status quo (cash, borrowing, asset sale)	Nothing — but it is free at the point of decision and needs no institutional trust	The real competitor

in the public materials of Avaneer, Synaptic or Change Healthcare. For them it would be a liability. For Obhoy it is the product.

Direct — domestic. No blockchain claims infrastructure is known to run in Bangladesh. Conventional insurtech vendors exist, but each serves one insurer, so none can solve cross-insurer duplication.

Indirect, and the real competitor. The status quo is cash, borrowing, asset sale and NGO community schemes. It is entrenched, free at the point of decision, and needs no trust in an institution. **Obhoy has to beat it on credibility, not only on price.**

B. What the consortium graveyard teaches

The useful precedents are not the survivors. They are one failure and one success, which together name the variable that mattered.

B3i was the best-resourced attempt at this exact class of infrastructure: a reinsurance consortium backed by twenty carriers including Swiss Re, Allianz, Munich Re, Zurich and SCOR. It raised about \$23m and filed for

insolvency in July 2022 — investors blamed demand, not the technology: “*we did not see the volumes in the demand*” [5]. It built industry-wide infrastructure, then went looking for transactions to put on it. **Insurwave** did the opposite: EY and Guardtime built it around one committed anchor client, A.P. Moller–Maersk, and went live in May 2018 covering more than 1,000 vessels from day one [6].

The lesson is structural, and it sets the roadmap in Section X. A claims-integrity network is a two-sided market, and two-sided markets do not start from breadth. Obhoy is therefore built so that *one insurer plus one aggregator in one upazila is a working deployment* — useful before it is complete. The multi-line design of Section IV-B says where the protocol can go, not where it starts. Building for four lines and launching into none is precisely how B3i failed.

C. Risk assessment and mitigation

Adoption risk — households do not believe the promise. This is the deepest risk and the reason Mechanism 4 exists. Where the honest current answer to “will they actually pay?” is 9.37% [1], only a published, externally anchored claims-paid ratio can answer it. We reduce it further by settling the first cohort fast and visibly, and by distributing through MFIs whose field relationships already carry trust.

Insurer and provider non-participation. Insurers may prefer opacity, and facilities profiting from inflated claims will resist attestation. We lead with the regulator, so joining carries a supervisory benefit, and make accreditation a condition of insured volume, starting with facilities already tied to the NGO partner. What the first insurer buys is not cross-insurer detection, which at $n = 1$ it cannot have: it joins for attestation before payment, inflation-proof settlement, provider accreditation history, and a published claims-paid record — one insurer plus one aggregator is already a viable pilot, and cross-insurer prevention is what a second insurer adds. To test the uniqueness invariant before a second real insurer exists, Phase 2 runs a second, non-underwriting InsurerMSP identity, held by IDRA or the academic node, submitting scripted duplicate claims against live events all year: Mechanism 1 under adversarial test on the production network, not only the negative-test suite of Appendix A. Phase 4 then tests economics, not mechanism.

Provider–verifier collusion. This is the most serious technical fraud route: two of the three classes colluding satisfies the quorum (Table III). We rotate field-verifier assignment so pairings do not stay stable. We run off-

chain anomaly detection on pairings against on-chain data, sample settled claims for audit, and raise the attestation requirement above a value threshold. **The risk is bounded and monitored, not removed** — the project’s most serious unsolved problem.

Identity aliasing — assumption A2 fails. A claimant enrolled under two national identities defeats the uniqueness invariant *before* chaincode sees it; chaincode cannot detect what it cannot resolve. This is the residual behind Table III’s strongest guarantee, and the honest price of the protocol: claims integrity becomes identity integrity. We enrol through the national ID service rather than self-declaration, alert on duplicate commitments across aggregators, and use biometric de-duplication where the MFI already runs it. The residual is bounded by the national identity system, outside the protocol — stated, not assumed away.

Actuarial risk — the benefit schedule is wrong. Too generous bankrupts the pool; too mean fails the household. We start conservative, reprice annually under council governance on published claims data, and reinsure tail risk behind a stop-loss layer and a disclosed pool-level cap for correlated catastrophic claims. A household can still face costs above the schedule — an honest trade, since cover that exists and pays beats comprehensive cover nobody can buy; tiering as the pool matures reduces it.

Remaining risks, briefly. Adverse selection and moral hazard: group-only enrolment, a waiting period, inpatient-only cover. Regulatory delay: IDRA joins from the start, not after building. Data protection: no single on-chain field is PHI, with metadata risk a DPIA question (Section V-D). Key loss: custodial keys with social recovery (Section V-J), whose residual is a trusted process re-entering. Chaincode defect: negative tests against the invariant, independent audit, staged upgrade, and a manual backstop that freezes state rather than mis-settling it.

The binding constraint is adoption, not cryptography. The hardest problems left in this project are institutional. The technology is the part we are most confident about.

XIII. CONCLUSION

Bangladesh has the medicine, the meteorology and the payment rails; what it lacks is a way for a buyer to trust an insurer’s promise, and an insurer to trust a claimed loss. Obhoy is a record no single party controls: opposed interests must attest before payment, and the result is published where it cannot be quietly edited

later. Its guarantees hold under four stated assumptions (Section IV-G) — sound identity resolution chief among them — and adoption, not cryptography, will decide whether this works. What it buys is the chance that a family facing surgery is not also facing the loss of everything they own.

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APPENDIX A

EVENT AND ENTITLEMENT INVARIANTS

The chaincode enforces the following over events e and entitlements c (Sections IV-C–V-C), alongside bound (1), holding only under A1–A4; chaincode cannot enforce A2 (Section IV-G).

$$\text{Open}(e) \wedge \text{Open}(e') \Rightarrow \text{Sub}(e) \neq \text{Sub}(e') \quad (4)$$

$$\text{Cont}(e', e) \Rightarrow \text{Sub}(e') = \text{Sub}(e) \wedge \neg \text{Setl}(e') \quad (5)$$

$$\text{Elig}(e) \Rightarrow |\{\text{cls}(a)\}_{a \in \text{Att}(e)}| \geq 2 \quad (6)$$

$$\text{Elig}(e) \Rightarrow \exists a \in \text{Att}(e) : \text{cls}(a) \neq \text{Pay}(e) \quad (7)$$

$$\text{Ev}(c) = \text{Ev}(c') \Rightarrow \text{Pol}(c) \neq \text{Pol}(c') \quad (8)$$

$$\text{Setl}(c) \Rightarrow \text{Elig}(\text{Ev}(c)) \wedge \text{Act}(c) \wedge \neg \text{Reop}(c) \quad (9)$$

Here Att, Sub, cls, Pay, Ev/Pol, Act are the attestation set, subject commitment, attester’s class, payee, event/policy and policy validity ($e \neq e', c \neq c'$). **The tests that matter most must not fail:** a transfer, a readmission and a second valid policy must all settle.